
Training Budget and Tokenization Change Which Ablation Conclusions Hold

Saghar Khalilpour

A Controlled Study on a Decoder-Only Transformer | September 2026

github.com/saqarkpr

Abstract

Hyperparameter ablations are usually run once, at one training budget, and reported as if the resulting ranking were a property of the architecture. We test that assumption directly by re-running an identical ablation suite — context length, depth, learning rate, weight tying — on the same decoder-only Transformer at two training budgets (500 and 5,000 iterations) and two tokenization granularities (character-level and byte-level BPE), with a measured per-configuration noise floor from repeated seeds. Two conclusions invert completely between conditions. A learning rate that outperforms the baseline by 55.9 perplexity points at 500 iterations underperforms it by 52.3 points (41σ) at 5,000. A depth ablation that is real and monotonic (22.8σ) at the character level is statistically indistinguishable from noise at the subword level, holding architecture and budget fixed. We report both reversals with their mechanism, and argue that any ablation table reported without a stated training budget, tokenization, and seed-noise estimate should be read as provisional.

1 Motivation

An architectural ablation — does depth help, does weight tying help, is this learning rate better than that one — is normally run once and reported as a fact about the architecture. Two assumptions are implicit in that practice: that the ranking observed at whatever training budget was used would hold at convergence, and that it would hold under a different tokenization of the same data. Neither assumption is obviously safe, and neither is usually tested, because testing it means re-running the same ablation twice under controlled conditions rather than once.

This report re-runs one ablation suite under exactly those two manipulations. The result is not a claim that any one architectural choice is good or bad; it is a measurement of how much an ablation ranking can move when the only things that change are how long training ran and what unit of text the model operates over.

2 Setup

The model is a decoder-only Transformer implemented from primitives — explicit causal multi-head self-attention, sinusoidal positional encoding, pre-norm residual blocks, GELU feed-forward layers — with no dependence on a higher-level modeling library. Two tokenizations of the same corpus (90/10 train/validation split of a ~ 1.1 M-character natural-language text) are studied: a 65-symbol character-level vocabulary, and a 2,000-token byte-level BPE vocabulary trained from scratch on the same corpus. The architecture is otherwise identical across both settings.

Baseline configuration. $d_{\text{model}} = 128$ (char) or 192 (BPE), 4 attention heads, 4 layers, context length 128, batch size 64, learning rate 3×10^{-4} , AdamW, gradient clipping at norm 1.0.

Noise floor. Three seeds at the baseline configuration establish a per-setting standard deviation in validation perplexity: $\sigma = 0.04$ (char, 5,000 iters) and $\sigma = 1.27$ (BPE, 5,000 iters). Every ablation effect below is reported in units of its own setting's σ , and an effect below 3σ is treated as not

distinguishable from noise regardless of its point estimate.

3 Ablation Suite

Four axes, one hyperparameter varied at a time from the baseline with everything else held fixed: context length ($\{32, 128, 256\}$), depth ($\{2, 4, 6\}$ layers), learning rate ($\{10^{-4}, 3 \times 10^{-4}, 10^{-3}\}$), and weight tying (embedding matrix shared with the output projection, or not). The BPE setting was run at both 500 and 5,000 iterations; the character setting was run at 5,000 iterations only, so all cross-budget comparisons below are within the BPE setting and all cross-tokenization comparisons are at the matched 5,000-iteration budget.

Axis	Config	Val. PPL	Δ vs. best	σ from best
Context	32	60.61	+2.91	2.3
	128 (best)	57.70	—	—
	256	59.66	+1.97	1.6
Depth	2 layers	57.75	+0.06	0.0
	4 layers (best)	57.70	—	—
	6 layers	60.91	+3.21	2.5
Learning rate	10^{-4}	72.70	+15.00	11.8
	3×10^{-4} (best)	57.70	—	—
	10^{-3}	110.02	+52.32	41.2
Weight tying	untied (best)	57.70	—	—
	tied	69.82	+12.12	9.6

Table 1: BPE setting, 5,000 iterations, $\sigma = 1.27$. Only weight tying and learning rate clear 3σ ; context and depth do not.

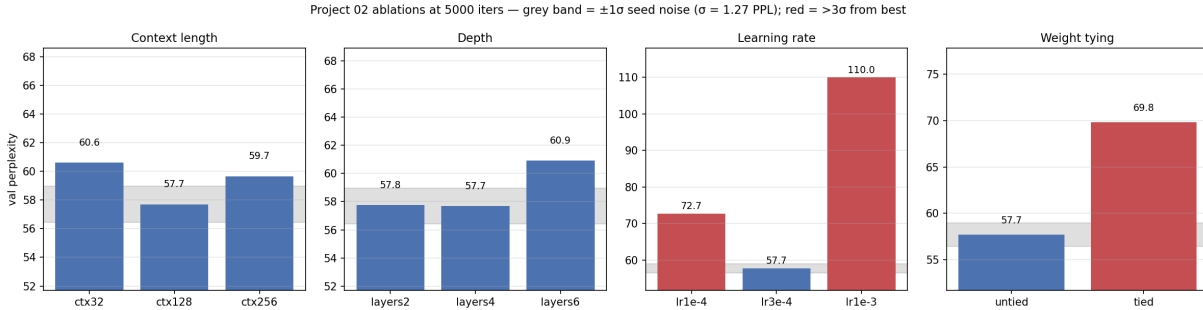


Figure 1: All four axes at 5,000 iterations. Grey band is $\pm 1\sigma$ around the best configuration in each panel; red bars sit outside it.

4 Finding 1: The Learning-Rate Ranking Inverts

At 500 iterations, $lr = 10^{-3}$ outperforms the baseline $lr = 3 \times 10^{-4}$ by 55.9 perplexity points. Reported alone, this supports raising the learning rate. At 5,000 iterations the same comparison reverses: $lr = 10^{-3}$ underperforms the baseline by 52.3 points, a 41.2σ effect (Table 1, Figure 2 left).

The $lr = 10^{-3}$ run reaches its best value, 56.50 perplexity, at iteration 2,250 — lower than any other configuration in the suite ever achieves — and then rises monotonically to 105.27 by iteration 5,000 while training loss continues to fall. The run is not under-trained; it has overfit, and the higher

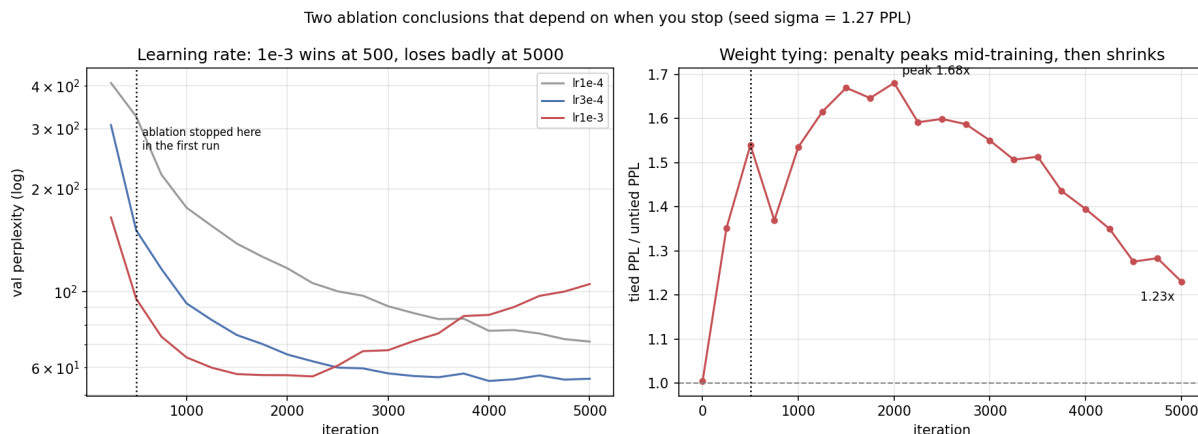


Figure 2: Left: validation perplexity by learning rate, log scale, with the 500-iteration stopping point marked — the rankings on either side of that line are opposite. Right: the weight-tying penalty (tied / untied perplexity ratio) across training, peaking at iteration $\sim 2,000$ before declining.

learning rate reached that regime faster than the baseline did. An ablation stopped at 500 iterations reports a ranking from before the crossover; one stopped at 5,000 reports a ranking from after it.

5 Finding 2: The Depth Conclusion Depends on the Tokenizer

At the character level, under an identical architecture and the same 5,000-iteration budget, depth is unambiguous: perplexity falls monotonically from 5.86 (2 layers) to 5.22 (4 layers) to 4.95 (6 layers) — 22.8σ and 6.7σ from the best configuration, against a measured $\sigma = 0.04$.

At the subword level, the same axis, same budget, same architecture gives 57.75 / 57.70 / 60.91 — the 2-and-4-layer configurations are indistinguishable (0.04σ apart) and 6 layers is nominally *worse*. Because both settings were run at the matched 5,000-iteration budget, training length is not the explanation here; the tokenizer is. BPE compresses the ~ 1.1 M-character corpus to ~ 390 K tokens, so at a matched iteration count the subword model has seen roughly a third of the effective data the character model has, and its 6-layer configuration bottoms out at iteration 4,000 before degrading — the signature of overfitting on a smaller effective dataset, not of depth being unhelpful in general.

The same architecture, ablated the same way, at the same budget, gives opposite conclusions about depth depending only on tokenization.

6 Finding 3: Weight Tying’s Penalty Shape Differs by Tokenization

Weight tying costs perplexity at both tokenizations, but the shape of that cost across training differs. At the character level the tied/untied ratio declines monotonically from its first measurement ($1.94\times$ at iteration 250) to $1.07\times$ at iteration 5,000. At the subword level the ratio *ris*es to a peak of $1.68\times$ at iteration $\sim 2,000$ before falling to $1.23\times$ by iteration 5,000, still declining when the run ends (Figure 2, right).

The direction predicted from the character-level result — that the tying penalty shrinks with more training — holds on both tokenizations. The specific shape of that shrinkage does not transfer: monotonic decay is a property of the character-level setting, not a general property of weight tying.

7 Discussion

Three results, one shared cause. An ablation ranking is a statement about a model at a particular point on its training trajectory, and that point is determined jointly by iteration count and by how much information the tokenization packs into each training step. Two axes (learning rate, depth) produced a ranking that inverted or vanished once that point changed; the third (weight tying) produced a direction that transferred but a magnitude profile that did not.

The practical implication is not that ablations are uninformative, but that an ablation table without a stated training budget, a stated tokenization, and a seed-based noise floor is a claim about one point in a larger space, not a claim about the architecture. All three are cheap to report and are rarely reported together.

8 Limitations

- Single corpus, single language, single random architecture family. Whether the specific crossover points (iteration $\sim 2,500$ for the learning-rate reversal, $\sim 2,000$ for the tying-penalty peak) generalize to other data or model scales is untested.
- Three seeds per configuration bounds the noise floor but does not pin it down precisely; σ itself has uncertainty that is not propagated into the σ -multiples reported above.
- Ablations vary one axis at a time from a single baseline point. Interaction effects between axes — for instance, whether the optimal learning rate shifts with depth — are out of scope here.
- The character-level run was not repeated at 500 iterations, so the depth finding’s budget-independence (Section 5) is inferred from a same-budget, cross-tokenization comparison rather than a full budget $\times$ tokenization grid.

9 Path Forward

The natural next experiment is a designed budget sweep — independent runs at 250, 500, 1k, 2.5k, 5k, and 10k iterations, each with its own seed study — reporting how many of a fixed set of ablation conclusions flip as budget increases, on more than one corpus and more than one tokenization. That would convert the two reversals reported here from case studies into a measured rate, which is the quantity a methodological claim of this kind actually needs.

10 Reproducibility

All code, the exact configurations behind every number in Table 1, the per-step training logs behind both figures, and the scripts that regenerate them from those logs are available at <https://github.com/saqarkpr/tokenization-ablation-study>. Every run uses a fixed, logged random seed; the noise floor in Section 2 is itself computed from three additional seeded runs at the baseline configuration, also included.